

Rev.1 active input: externally protected 2S 18650 pack + external BMS only. J4 is BAT+/BAT- 6.0...8.4V; no midpoint, charger or USB-C input.

BAT+ -> F1 1812L200/16 -> BAT_FUSED -> Q1 DMP3130LQ-7 reverse-polarity MOSFET -> BUCK_IN -> U1 TPS62133RGT -> 5V_SYS.

D3 SMBJ10CA is a bidirectional input transient clamp at BAT_FUSED. No on-board charger/BMS is duplicated; no main-board switch is fitted in Rev.1.

Rev.1 policy: disconnect battery or turn its external BMS output off before connecting the removable DevKit USB-C to a computer for programming.

TPS62133: PVIN/AVIN/EN=BUCK_IN, SW1-3 -> L1 2.2uH -> 5V_SYS, VOS/FSW=5V_SYS, FB/DEF/grounds/EP=GND, SS/TR=C4 to GND; PG NC.

J3 is universal E220-T22D: fit one E220-400T22D OR E220-900T22D. M0/M1 10k pulldowns; OLED VCC/pull-ups signal-only NC/DNP.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, 5V) -> WS2812B-V5 DIN. No PCB footprint/layout work is included.

TP1 BAT_PLUS, TP2 GND, TP3 BUCK_IN, TP4 5V_SYS, TP5 E220_VCC(5V_SYS), TP6 M0, TP7 M1, TP8 AUX, TP9 RXD, TP10 TXD.

